

ARNAV SINGH

Research-Oriented ML Engineer | Model Evaluation • Imbalanced Learning • LLM Fine-Tuning • Explainable AI

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RESEARCH SUMMARY

Final-year B.Tech Computer Engineering student (CGPA 8.45) who treats every ML project as an experiment: frame a question, set a baseline, test the assumption, and report what held up, including what didn't. Ran a LoRA-rank ablation (ranks 4 to 64) when fine-tuning Mistral-7B, chose evaluation metrics by the real cost of errors on a dataset with under 1% positives (AUC-PR 0.9752), and validated churn signals on 540K+ transactions with chi-square and t-tests before recommending action. Builds the experiment through to a reproducible, deployed system with MLflow, Docker and FastAPI. Currently applying correlation-based validation to multi-year electoral data as a Data Analyst Intern.

RESEARCH INTERESTS AND METHODS

Interests: Model evaluation and benchmarking • Learning from imbalanced data • Parameter-efficient LLM fine-tuning • Explainable and trustworthy ML • Uncertainty quantification

- **Problem formulation:** start from the decision and the cost of a wrong answer, then pick the metric (Fraud Detection, Fake ID Screening).
- **Controlled experiments and ablations:** change one factor at a time; benchmark quality, memory and latency (LegalSimplify, LoRA ranks 4–64).
- **Statistical inference:** chi-square, t-tests and correlation tests before claiming an effect (Churn & Cohort, electoral data).
- **Evaluation design:** precision-recall, AUC-PR, per-class F1, perplexity, BERTScore; documented failure modes (Fraud, Legal Classifier, LegalSimplify).
- **Interpretability:** global and per-prediction SHAP, translated into rules non-experts can check (Fraud Detection).
- **Forecasting and simulation:** Elo ratings with Monte Carlo simulation; predictions published before the event, misses included (FIFA 2026 Predictor).
- **Reproducibility:** experiment tracking in MLflow and Weights & Biases; Docker images for every served model (Fraud, Legal Classifier).

RESEARCH PROJECTS

LegalSimplify — Parameter-Efficient Fine-Tuning of Mistral-7B for Legal Text | *Hugging Face, PEFT, QLoRA, BitsAndBytes* | [GitHub](#)

- **Question:** How small can a LoRA adapter be before plain-language quality on legal clauses drops, and what does each rank cost?
- **Method:** Fine-tuned Mistral-7B-Instruct on a custom legal dataset with LoRA and 4-bit quantisation; ran an ablation over ranks 4, 8, 16, 32 and 64, benchmarking perplexity, inference latency and task quality.
- **Finding:** Rank 16 gave the best quality-to-cost trade-off for the legal domain; 4-bit QLoRA cut training memory by about 75%.

Credit Card Fraud Detection — Evaluation Under Extreme Class Imbalance | *XGBoost, SMOTE, SHAP, MLflow, FastAPI, Docker* | [Live Demo](#) · [GitHub](#)

- **Question:** With fraud under 1% of 284K transactions, which metric and decision threshold reflect the real cost of missed fraud versus false alarms?
- **Method:** Compared resampling and model choices tracked in MLflow; evaluated on AUC-PR instead of accuracy; set the threshold with a cost-based analysis.
- **Finding:** AUC-PR 0.9752, 87.5% recall at a 0.15% false-alarm rate, and an estimated £5,338 in prevented losses against a no-detection baseline.
- **Explainability:** SHAP surfaced the top 5 behavioural risk drivers, translated into candidate alert rules. Deployed as a Dockerised REST API with a real-time inference endpoint.

Customer Churn and Cohort Retention — Statistical Validation of Churn Signals | *SQL, Python, statsmodels, Power BI* | [GitHub](#)

- **Question:** When do customers leave, and is the drop real or noise?
- **Method:** Built acquisition cohorts and monthly retention curves from 540K+ transactions with SQL CTEs and window functions; tested churn signals across cohorts with chi-square and t-tests.
- **Finding:** Month 3 is the critical churn window and the main revenue risk; quantified revenue at risk per cohort and modelled recovery potential.

Legal Query Classifier — Failure-Mode Analysis of a Fine-Tuned BERT | *BERT, PyTorch, FastAPI, Docker* | [GitHub](#)

- Fine-tuned BERT for multi-task legal text classification (90%+ accuracy), handling class imbalance with weighted loss.
- Reported per-class precision, recall and F1, and documented known failure modes for monitoring and future iteration.

AI-Based Fake Identity and Document Screening | *OpenCV, EasyOCR/Tesseract, DeepFace/InsightFace, Streamlit* | [GitHub](#)

- Designed a five-stage pipeline (OCR → validation → tamper detection → face verification → risk scoring), each stage an independently testable component.
- Combined ICAO 9303 MRZ check-digit parsing, Error Level Analysis and face-embedding similarity into a tunable weighted risk score returning real-time PASS/FLAG verdicts.

FIFA World Cup 2026 Prediction Model | *Python, Elo ratings, Monte Carlo simulation* | [Live](#)

- Built an Elo-based rating model with Monte Carlo tournament simulation and published predictions before kick-off.
- Called 4 of 5 knockout matches correctly (80%), with the misses published alongside the hits.

RESEARCH EXPERIENCE

Data Analyst Intern — Jambudweep Shodh Kendra Pvt Ltd

Aug 2026 – Present

- Analyse multi-year electoral datasets across constituency, Lok Sabha and Rajya Sabha elections to identify voting-pattern shifts and long-term trends.
- Clean and structure raw election records into analysis-ready datasets, then build trend visualisations and summary reports for decision-makers.
- Validate predicted trends against historical results using statistical correlation tests, improving the reliability of reported insights.

Retina AI Hackathon — 3rd Place · Student dropout-risk prediction

2026

- Built a dropout-risk model fusing three signal types: grades, attendance trends and counsellor notes.

TECHNICAL SKILLS

- **Data and analysis:** Python (Pandas, NumPy), SQL (CTEs, window functions; MySQL, PostgreSQL), EDA, feature engineering, cohort analysis
- **Statistics:** Hypothesis testing (chi-square, t-test), correlation and regression (statsmodels), A/B testing, conformal prediction, Monte Carlo simulation
- **Modelling:** Scikit-learn, XGBoost, PyTorch, imbalanced learning (SMOTE, ADASYN, weighted loss), CNNs
- **LLMs and NLP:** Hugging Face Transformers, PEFT (LoRA, QLoRA), BitsAndBytes 4-bit quantisation, Mistral-7B, BERT, FAISS, prompt engineering
- **Evaluation and interpretability:** AUC-PR, precision/recall/F1, perplexity, BERTScore, latency benchmarking, ablation studies, hyperparameter tuning, SHAP
- **Reproducibility and deployment:** MLflow, Weights & Biases, Docker, FastAPI, Streamlit, Render, AWS SageMaker (basics), Git
- **Communication:** Power BI (DAX), Tableau, Matplotlib, Seaborn, stakeholder reporting

EDUCATION

B.Tech, Computer Engineering — ABES Engineering College (AKTU)

2023 – 2027

CGPA 8.45 · Relevant coursework: Machine Learning, Probability & Statistics, Data Structures & Algorithms, DBMS, Operating Systems

Class XII: 81% · Class X: 92.4% — St. Thomas School, Indirapuram (CBSE)

ACHIEVEMENTS AND CERTIFICATIONS

- 3rd Place, Retina AI Hackathon 2026; 2nd Place, ECell Eureka Competition Zonals (IIT Bombay E-Cell)
- 2025 Google Solution Challenge Participant; Google 5-Day AI Agents Intensive Contributor
- 300+ DSA problems solved on LeetCode and CodeChef; CodeChef rating 1663 (3★)
- Certifications: Data Science and ML Fundamentals; Master Regression & Prediction with Pandas and Python; SQL for Data Analytics; IBM SkillsBuild Data Fundamentals

LEADERSHIP

- **Growth & Content Strategist, Aasira – STEM Beyond Labels (Jul 2024 – Feb 2026):** ran weekly engagement analysis to guide content, lifting engagement 35%.
- **Volunteer, GDGoC (Oct 2024 – Sep 2025):** used reach and timing data to optimise campaigns, raising event attendance 40%.